Meeting 4 of 7 · October 27, 2026

NOISE AND MITIGATION

Noise breaks calibration; mitigation repairs it

Noise models · calibration vs discrimination · readout mitigation · zero-noise extrapolation · resource-matched tests

Deeper analysis of Phases 3B and 4 (completed Oct 7, 2026). Noise damaged calibration and threshold decisions, not discrimination. Mitigation repaired calibration; three times the shots did not.

Mayank Kumar Pokhriyal · **Tushar Poonia**
M.S. in Artificial Intelligence capstone · Katz School of Science and Health, Yeshiva University
Proposed split: Tushar (slides 1–3) · Mayank (slides 4–5) · about 5 minutes



How we simulated noise

Depolarizing

after every 1- and 2-qubit gate

low · medium · high

two-qubit error 0.005 · 0.010 · 0.020

Thermal relaxation

during 35 ns and 300 ns gates

low · medium · high

T_1 200 · 100 · 50 µs

Readout confusion

bit flips at measurement

low · medium · high

flip probability 0.005 · 0.02 · 0.05

12 anchor conditions
3 sources × 3 severities + 3 combined

→

Exact simulation
Qiskit Aer density matrices

→

Finite shots
1,024 shots × 20 replicates

→

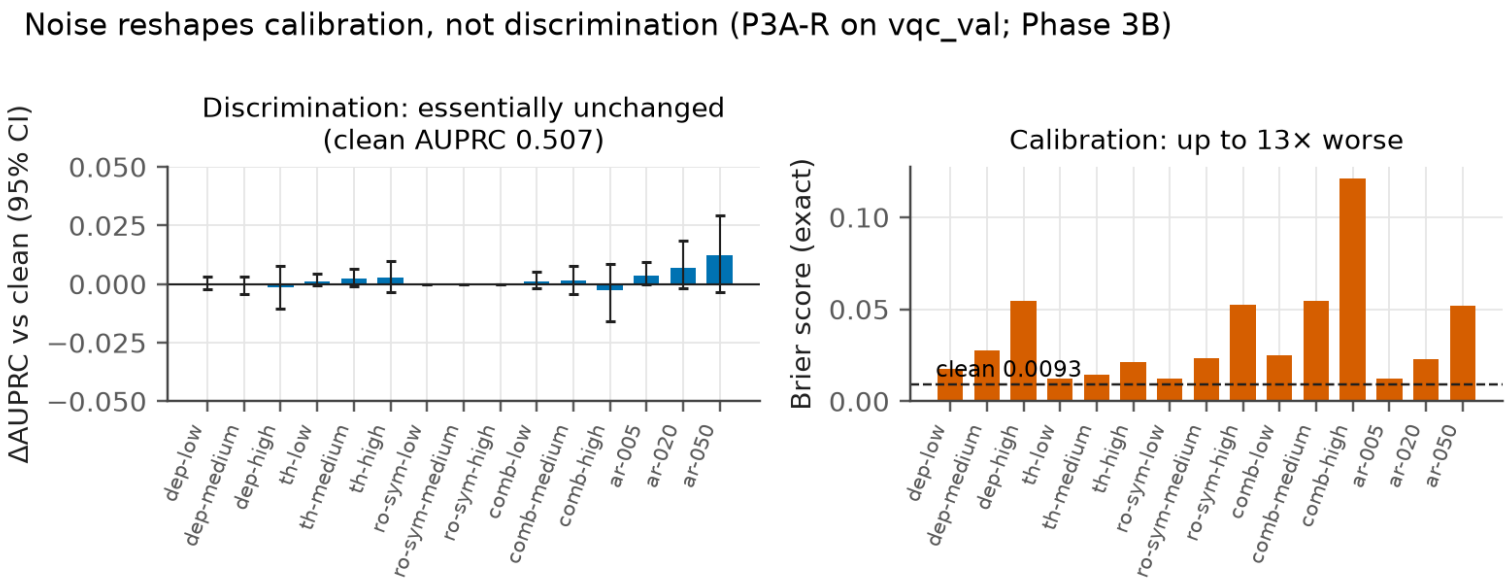
Negative control passed
symmetric readout: ranking metrics unchanged (Δ = 0)

Plus 3 synthetic asymmetric-readout stress profiles (mean error 0.005–0.05). These are an assumption, not device calibration data.

Exact simulation at the registered anchor severities: effects come from the noise model, not from sampling.

Source: configs/noise/anchors_rr_v1.yaml, p3b_noise_matrix.csv, p3b_negative_control.csv

Noise damages calibration, not discrimination



≤ 0.003

change in AUPRC; ≥ 46 of 47 top frauds kept

0.0093 → 0.121

Brier score, combined-high noise (13×)

0.083 → 0.346

expected calibration error

0.53 → 0.03

precision at the frozen alert threshold

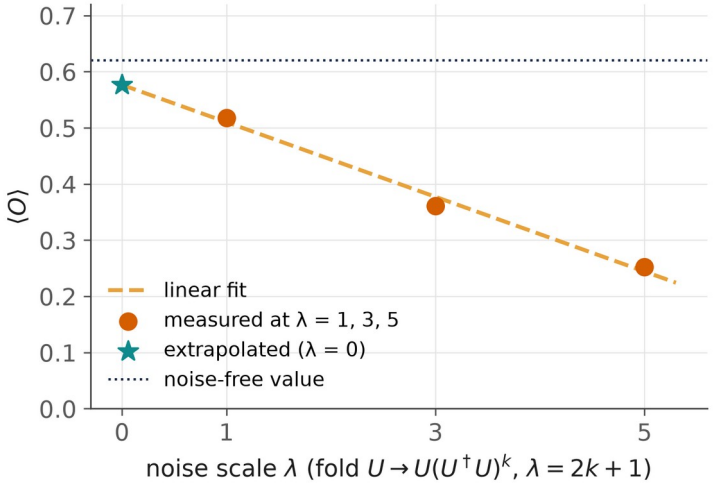
Mitigation's job here is calibration and threshold decisions, not discrimination.

Source: p3b_exact_metrics.csv (P3A-R, exact simulation), p4_exact_diagnostics.csv

Two mitigation tools, very different costs

Readout mitigation (A1)

- 2 calibration circuits per batch: |0000⟩ and |1111⟩
- per-qubit confusion matrices from counts → tensored inverse
- project negative quasi-probabilities onto the simplex



Illustrative values (not project data).

ZNE (A2)

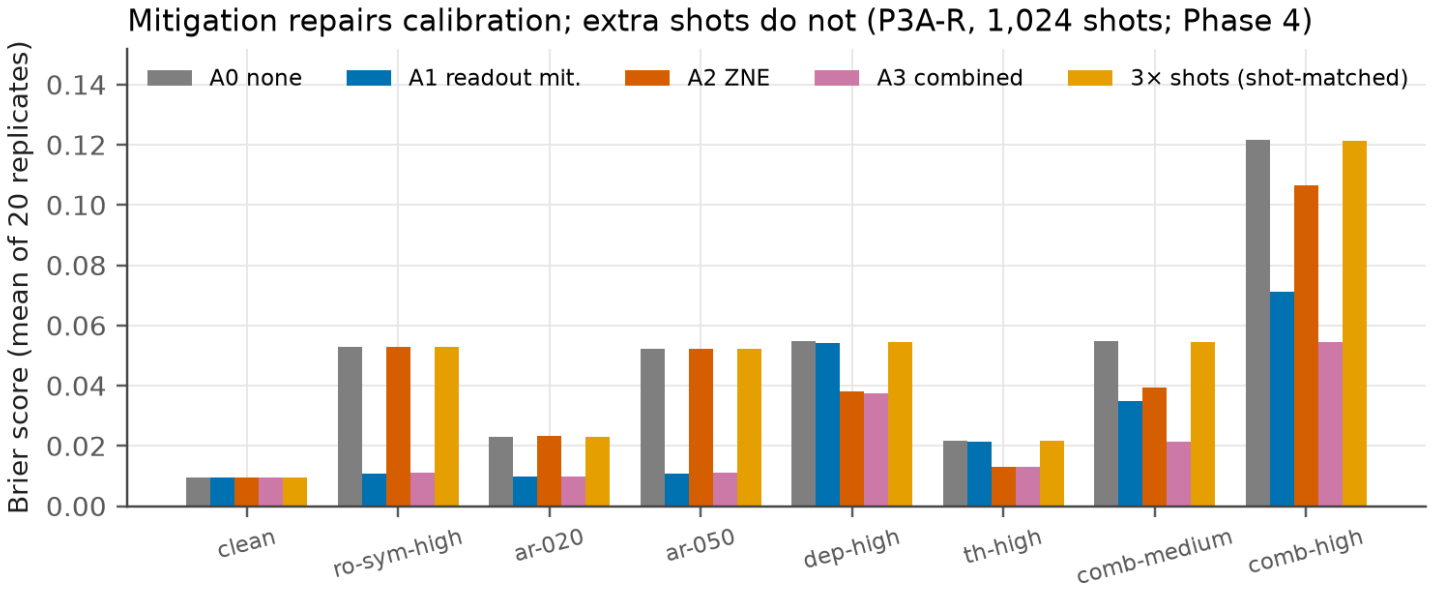
- fold the circuit: λ = 1, 3, 5
- linear fit to λ = 0
- A3 = readout mitigation at every scale, then ZNE

Action	What it does	Shots per inference	Two-qubit-gate exposure
A0	nothing	S	1×
A1	readout mitigation	S + 2S/N	1×
A4	three times the shots, unmitigated	3S	3×
A2	zero-noise extrapolation	3S	9×
A3	readout mitigation + ZNE	3S + 2S/N	9×

S = shots per circuit · N = transactions per batch · the extrapolation curve is illustrative

Source: configs/mitigation/*.yaml, p4_costs.csv

Mitigation repairs calibration; extra shots do not



Brier (1,024 shots)	None	Best	Action	3× shots
Readout noise, high	0.053	**0.011**	A1 readout	0.053
Depolarizing, high	0.055	**0.038**	A2 ZNE	0.055
Combined, high	0.122	**0.054**	A3 both	0.121

What it means

- Three times the shots: Brier unchanged (shots cut variance, not bias)
- No action improved AUPRC over its resource-matched comparator
- Linear ZNE recovered 13–70% of the exact Brier gap
- The best action depended on the noise type

Mitigation removes a bias that extra shots cannot. Which action pays depends on the noise type, which is the Nov 10 topic.

Source: p4_summary.csv (1,024 shots), p4_resource_matched.csv, p4_best_action.csv